

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0257

Barre 100 kV
September 16, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0257. The Interconnection Customer proposes a 203.4 MW solar photovoltaic facility interconnecting at the Barre 100 kV substation in the Indiana area. The request was received on July 31, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 139 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,941,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Indego / Columbia Hills 100 kV is overloaded to 137.3% of its 148 MVA emergency rating for the loss of the Waynedale / Monfort 230 kV Line. The project contributes 17.1% DFAX (34.8 MW) to this overload.

The Columbia Hills / Portage Uppco Substation 100 kV is overloaded to 135.9% of its 123 MVA emergency rating for the loss of the Waynedale / Monfort Substation 230 kV. The project contributes 15.8% DFAX (32.2 MW) to this overload.

The Westboro / Sapony 100 kV is overloaded to 131.9% of its 127 MVA emergency rating for the loss of the Barre / Raymond F. Barker 100 kV. The project contributes 100.0% DFAX (203.4 MW) to this overload.

For the loss of Barre / Raymond F. Barker 100 kV L, the Brandford / Westboro 100 kV Line loads to 130.3% of its 122 MVA rating. Project DFAX: 100.0%.

The Brandford / Limon Wind 2 100 kV is overloaded to 128.1% of its 111 MVA emergency rating for the loss of the Barre / Raymond F. Barker 100 kV line. The project contributes 100.0% DFAX (203.4 MW) to this overload.

Sapony Substation / Stagecoach 100 kV line: 126.8% loading (141 MVA) following the outage of the Raymond F. Barker / Barre 100 kV; DFAX = 100.0%.

The Segs 5 / Aep Pettit Avenue 100 kV is overloaded to 126.2% of its 138 MVA emergency rating for the loss of the Waynedale / State Street 230 kV. The project contributes 6.5% DFAX (13.2 MW) to this overload.

Waynedale / Indego 100 kV: 125.6% loading (134 MVA) following the outage of the Monfort / Waynedale 230 kV; DFAX = 17.4%.

The Peavine / Hohenwald 100 kV is overloaded to 124.6% of its 120 MVA emergency rating for the loss of the Waynedale / State Street 230 kV. The project contributes 6.7% DFAX (13.7 MW) to this overload.

For the loss of Waynedale / Monfort 230 kV, the Hart / Portage Uppco 100 kV loads to 122.7% of its 126 MVA rating. Project DFAX: 10.3%.

Aep Pettit Avenue / Waynedale 100 kV Circuit 2: 120.7% loading (172 MVA) following the outage of the Waynedale / State Street 230 kV Line; DFAX = 6.6%.

For the loss of Waynedale / Monfort 230 kV, the Southport / Hart Substation 100 kV line loads to 116.7% of its 117 MVA rating. Project DFAX: 10.3%.

For the loss of Waynedale Substation / Monfort 230 kV, the Illinois Rd / Southport 100 kV loads to 115.5% of its 102 MVA rating. Project DFAX: 10.3%.

The Barre / Raymond F. Barker 100 kV L is overloaded to 113.4% of its 157 MVA emergency rating for the loss of the Sapony / Stagecoach 100 kV. The project contributes 100.0% DFAX (203.4 MW) to this overload.

For the loss of Waynedale Substation / State St Substation 230 kV, the Industrial Park / Spy Run 100 kV L loads to 113.2% of its 118 MVA rating. Project DFAX: 5.4%.

The Waynedale / Aep Pettit Avenue 100 kV is overloaded to 113.0% of its 183 MVA emergency rating for the loss of the Waynedale / State St 230 kV. The project contributes 6.6% DFAX (13.4 MW) to this overload.

The Meridian / Limon Wind 2 Substation 100 kV L is overloaded to 108.9% of its 117 MVA emergency rating for the loss of the Raymond F. Barker / Barre 100 kV. The project contributes 100.0% DFAX (203.4 MW) to this overload.

For the loss of High Springs / Barre 100 kV, the Barre / Stagecoach 100 kV line loads to 104.8% of its 122 MVA rating. Project DFAX: 26.9%.

The High Springs / Barre 100 kV line is overloaded to 102.6% of its 127 MVA emergency rating for the loss of the Barre / Stagecoach 100 kV. The project contributes 27.3% DFAX (55.5 MW) to this overload.

Illinois Road / Hebron Rosemeade 100 kV L: 101.6% loading (109 MVA) following the outage of the Monfort / Waynedale 230 kV; DFAX = 10.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Columbia Hills / Indego 100 kV	Rebuild line with higher-capacity conductor	\$2,312,000	60% (\$1,386,000)
Columbia Hills / Portage Uppco 100 kV line	Rebuild line with higher-capacity conductor	\$3,161,000	66% (\$2,075,000)
Westboro / Sapony 100 kV	Rebuild line with higher-capacity conductor	\$4,860,000	100% (\$4,860,000)
Brandford / Westboro 100 kV Line	Rebuild line with higher-capacity conductor	\$4,094,000	100% (\$4,094,000)
Limon Wind 2 / Brandford 100 kV	Rebuild line with higher-capacity conductor	\$9,542,000	100% (\$9,542,000)
Sapony / Stagecoach 100 kV	Rebuild line with higher-capacity conductor	\$11,082,000	100% (\$11,082,000)
Segs 5 / Aep Pettit Avenue 100 kV	Rebuild line with higher-capacity conductor	\$2,610,000	51% (\$1,329,000)
Waynedale / Indego 100 kV	Rebuild line with higher-capacity conductor	\$3,230,000	60% (\$1,950,000)
Peavine / Hohenwald 100 kV	Rebuild line with higher-capacity conductor	\$1,736,000	55% (\$956,000)
Hart / Portage Uppco 100 kV	Rebuild line with higher-capacity conductor	\$3,642,000	65% (\$2,359,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aep Pettit Avenue / Waynedale 100 kV Circuit 2	Rebuild line with higher-capacity conductor	\$2,313,000	50% (\$1,148,000)
Southport / Hart 100 kV	Rebuild line with higher-capacity conductor	\$2,963,000	65% (\$1,919,000)
Illinois Rd / Southport 100 kV	Rebuild line with higher-capacity conductor	\$3,579,000	65% (\$2,318,000)
Barre / Raymond F. Barker 100 kV L	Rebuild line with higher-capacity conductor	\$9,914,000	100% (\$9,914,000)
Spy Run Substation / Industrial Park 100 kV	Rebuild line with higher-capacity conductor	\$1,432,000	42% (\$608,000)
Waynedale / Aep Pettit Avenue 100 kV	Rebuild line with higher-capacity conductor	\$2,072,000	50% (\$1,028,000)
Limon Wind 2 / Meridian 100 kV	Reconductor line	\$1,889,000	100% (\$1,889,000)
Barre / Stagecoach 100 kV	Reconductor line	\$2,824,000	100% (\$2,824,000)
High Springs / Barre 100 kV	Reconductor line	\$1,199,000	100% (\$1,199,000)
Illinois Road / Hebron Rosemeade 100 kV L	Reconductor line	\$714,000	65% (\$461,000)
	Total allocated to project		\$62,941,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.